As mentioned, there are several xG models made by several different people. However, the underlying premise in each remains the same – that of arriving at an “expected” number of goals for a team given details about the shots it took. The following discussion focuses on the xG model devised by Michael Caley, Analytics Editor for Howler magazine – a US publication about football (soccer, for them). It is by no means perfect, but then again, no predictive model ever has been.

Caley’s xG method considers the following factors in order to calculate the probability of scoring a goal from each shot.

Shot location: it’s found that as you go farther away from goal, the likelihood of scoring decreases exponentially with distance. Of course, the angle from which the shot is taken is accounted for as well, with an “adjusted distance” metric calculated as the actual distance from goal divided by the angle raise to the power 1.32.

Shot type and assist type: For various combinations of these two, the exponential decay rate is empirically adjusted. For example, a header from a cross will become difficult to score very quickly as you move away from goal.

Speed of attack: a linear relationship was found between the speed of attack and the likelihood of a goal – since a fast break usually happens when the opposition defence is out of position.

Other factors like whether the shot was from a direct freekick or a rebound – which greatly decrease and increase the likelihood of a goal respectively – are accounted for similarly in the exponentially decreasing function.

Finally, all these values corresponding to each shot taken are added up, yielding a single value – the expected goals.

However, this system has some important shortcomings that yield inaccurate results for what would be considered boundary conditions or edge cases.

KEEPING TRACK OF WATER LOSS

The Kerala Water Authority (KWA) has deployed IBM's Analytics and Mobility solution to analyze, track and manage water distribution in Thiruvananthapuram. The system regularly monitor and issue alerts using sensors and intelligent meters and the overall revenue collection improved to around 20%.